



Identification of health-related needs: The needs examination, evaluation and dissemination (NEED) assessment framework

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ABSTRACT

Innovation in healthcare is highly supply-driven, leading to underinvestment in less profitable health areas, unmet needs and inefficient use of public resources. This study proposes a framework to structure the identification of health-related patient and societal needs across different health conditions, supporting the evolution towards needs-driven healthcare policy and innovation. The Needs Examination, Evaluation and Dissemination (NEED) assessment framework was based on the results of two systematic literature reviews and stakeholder and expert consultation. The first review concerned patient needs criteria and the second concerned societal needs criteria. Relevant publications in Ovid Medline® or Embase®, in English, French or Dutch were included. The NEED framework addresses patient, societal, and future needs across health, healthcare, and social domains. Patient-level needs encompass five criteria in the health domain (e.g., impact on quality of life), four in the healthcare domain (e.g., burden of treatment) and four in the social domain (e.g., impact on social life). Societal needs encompass four health needs criteria (e.g., transmissibility), two healthcare needs criteria (e.g., value for money) and two social needs criteria (e.g., productivity losses). Future needs criteria (2) consider future burden of disease and economic burden. Equity is recognised as a transverse dimension, requiring unmet needs data disaggregated by population sub-groups. Each criterion is associated with one or more measurable indicators. This framework represents an important first step towards a more needs-driven healthcare policy and innovation landscape.

1. Background

The current health innovation ecosystem, primarily supply-driven, inadequately addresses high-priority unmet needs, leading to significant gaps in areas like mental health and antimicrobial resistance [1]. Transitioning towards needs-driven healthcare innovation and policy requires, as a first step, identifying and prioritizing of unmet needs across diseases. This should then be followed by the implementation of financial incentives or the allocation of funding to support investments in the identified unmet needs.

Currently, pharmaceutical research and development (R&D) tends to prioritize profitability over addressing pressing health concerns, leading to limited investment in areas of greatest need [2–4]. Additionally, regulatory processes focus on evaluating individual technologies rather than considering broader societal health needs [5], resulting in

inefficiencies and inequities in resource allocation. The overwhelming emphasis on drug development overshadows other health technologies and interventions such as preventive measures and cross-sector interventions targeting wider social and commercial health determinants.

Previous literature on bridging the gap between public health needs and R&D in the health sector has established pharmaceutical R&D agendas and lists of health conditions with a possible drug development gaps or high-burden, under-researched medical conditions [6–8]. However, a comprehensive framework to identify disease-specific unmet needs had not been developed until now.

European initiatives such as the Priority Medicines scheme PRIME [9] conditional marketing authorization [10], and accelerated assessment [11], as well as national initiatives like Belgium's compassionate use and (unmet) medical need programs [12,13], aim to support the development of medicines targeting unmet medical needs. However, the

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concept of unmet medical needs also used in the revision of the European Union (EU) pharmaceutical legislation, lacks a clear definition and has various interpretations across stakeholders in the healthcare sector [14–16].

The Belgian Health Care Knowledge Centre (KCE) has a history of researching the consideration of patients' and societal needs in health policy, including the development of a multi-criteria decision analysis approach for medical needs appraisal and of a methodology for identifying patient needs [17,18]. In the same vein, the Needs Examination, Evaluation and Dissemination (NEED) project was launched by KCE and Sciensano [19]. The goal of NEED is to establish an evidence database on unmet health-related needs of patients and society across various conditions, according to standardised procedures to ensure scientific quality.

This study, part of the NEED project, aims to develop the NEED assessment framework to structure the identification and measurement of unmet health-related patient and societal needs. The framework focuses on criteria defining health-related patient and societal needs and identifies potential indicators to evaluate how well these needs are currently met. Indicators include those requiring primary data collection. Data on the indicators can be used for different priority-setting purposes, such as research funding, reimbursement decisions, and provision of R&D incentives, by various health sector actors, including policymakers, researchers, patient representatives, or the industry. The appraisal of the relative importance of the different criteria is, however, beyond the scope of this study and should be performed by policymakers (Fig. 1).

1.1. Definitions: needs and unmet needs

The concept of “need” is used in different fields, such as economics, psychology, sociology, healthcare and marketing, to understand human behaviours and to prioritise interventions.

In the literature, the concept of need for healthcare has been approached from various perspectives. Among the most common approaches are the humanitarian and realistic theories, which provide distinct views on healthcare needs. In the humanitarian theory, the presence and extent of a healthcare need are directly correlated with the level of ill-health in a population [21]. The realistic theory argues that a healthcare need exists only when there is an evidence-based intervention available that can improve the outcomes of a disease. This implies that not every health need necessarily leads to a healthcare need, particularly when effective interventions are lacking or solutions lie outside the scope of health systems [20]. In addition, there is no need for healthcare if there exists a more resource-efficient intervention that achieves the same outcome.

The study's definition of need incorporates aspects from both theories. While we acknowledge that needs may arise even if effective interventions exist, we also consider that needs arise because an effective intervention does not (yet) exist or falls outside the health system's

scope. Needs like in the humanitarian theory are correlated with the burden of disease in a population.

We therefore consider the broader concept of “health-related needs”, defined as the specific needs associated with a particular health condition's trajectory, from prevention to diagnosis, treatment, and rehabilitation or palliative care. These needs encompass the needs of patient and those of the broader society. Health-related needs can be met by targeted healthcare interventions, health system-wide interventions, social services, or wider policies that have an impact on health.

This study built mainly on approaches by Santana et al. [22] and Incenti et al. [23], to define “unmet needs”. An unmet need arises when the available offer or supply of health-related interventions does not, or not completely or adequately, meet existing health-related needs [23]. Thus, needs can be expressed on a continuum ranging from ‘completely unmet’ (when no effective intervention is available) to ‘completely met’ (when current interventions effectively address the needs), and are dynamic over time [23].

2. Methods

The framework was developed on the basis of two literature reviews, one to identify potential patient needs criteria and another to identify potential societal needs criteria, complemented by stakeholder consultations.

For the two reviews, the screening, selection and data collection processes were guided by the PRISMA guidelines and were synthesized in multiple steps (see Figure A 1 and A3 in supplementary material). Rayyan software was used to facilitate the screening and selection of articles and EndNote was used as reference manager software.

2.1. Methods to identify patient needs criteria

To identify tools and methods for measuring patient needs, we updated the systematic review by Maertens de Noordhout et al. (2022) [18]. A systematic search was conducted in the electronic databases Cochrane, Embase®, and Ovid Medline® using logical combinations of keywords. A detailed list of all keywords and logical combinations per database is available in Table A1 of the supplementary material. All types of studies were included (see Table A1, supplementary material). Articles were included if they (1) measured patient health-related needs in any patient population, (2) provided evidence from Western countries (EU, UK, Iceland, Liechtenstein, Norway, Switzerland, United States, Canada, Australia, and New Zealand), (3) were published between December 2020 and January 2023, and (4) were in English, French, or Dutch.

After identifying the relevant literature, duplicates were removed, and a two-stage screening was performed. Selection criteria were used to check eligibility (see Table A7 in the supplementary material). The screening process involved two stages: title and abstract review by two team members (C.M. and L.K.), followed by a full-text review by one

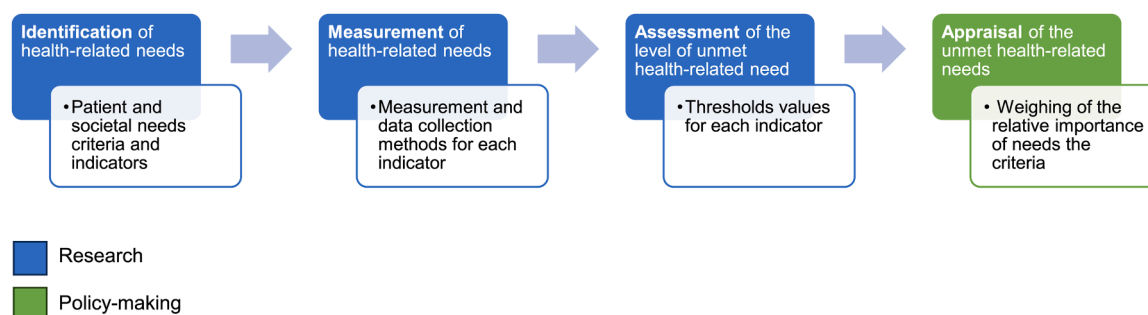


Fig. 1. The different processes in the Needs Examination, Evaluation and Dissemination (NEED) project. Note: the methods to identify threshold values and weights for the relative impact of the needs criteria are beyond the scope of this study.

team member (L.K.). Disagreements were resolved by consensus. Data extraction focused on the study's objectives, population, data collection methods, sample description, tools used to assess patient needs, and analysis type. Patient needs domains and specific criteria identified in each tool were listed (Table A5, supplementary material). These new domains and criteria were integrated with existing ones from the generic questionnaire developed in the previous KCE study [18] and validated by a Delphi panel of patient representatives.

2.2. Methods to identify societal needs criteria

For the identification of criteria to measure societal needs, a parallel systematic literature review was undertaken. Given the lack of standardized tools to measure societal needs, the focus shifted to identifying criteria applicable to specific health conditions and consistent with a needs-driven approach. Societal needs were defined as the consequences of a health condition on segments of society not directly afflicted by the condition. Searches were conducted in the electronic databases Medline® and Embase® using logical combinations of keywords. A detailed list of all keywords and logical combinations per database is available in Table A2 of the supplementary material. All types of studies were included if they (1) identified societal health-related needs in the general population or health system, (2) provided evidence from one or more European countries, (3) were published between January 2012 and January 2023, and (4) were in English, French, or Dutch.

After identifying the relevant literature, duplicates were removed, and a two-stage screening was performed. Selection criteria were used to check eligibility (see Table A8 in the supplementary material).

Two team members (M.L. and C.M.) conducted a two-stage screening: first, by title and abstract, then by full text (M.L. and C.M. for ten publications). Disagreements were resolved through consensus among the two reviewers. Information on the study's aim, target population, characteristics, and potential criteria or indicators for societal needs were extracted by one team member (M.L.) using a predefined data extraction form in Microsoft Excel. Identified criteria were listed, grouped thematically into overarching domains, and refined through expert reviews (Table A6, supplementary material). Domains or criteria not meeting four selection criteria were excluded: (1) measurability for a specific health condition; (2) measurability at the societal level, not solely at the patient level; (3) measured independently of health interventions not yet on the market; and (4) minimal overlap with patient needs domains/criteria.

In April 2023, a stakeholder consultation meeting was organised to discuss the pre-selected societal needs criteria list. Participants included representatives from Belgian federal administrations, sickness funds, the pharmaceutical industry, patient associations, the Belgian Cancer Registry, and academia (Table A3, supplementary material). Stakeholders designated representatives to participate in the consultation. Prior to the meeting, they received the list of criteria and were asked to rate each as high, medium, or low relevance, and as essential or nice-to-have for identifying societal needs. Stakeholders could also comment on missing domains or criteria. When >60 % of participants rated a criterion as having high relevance, or when >50 % of respondents rated a criterion as essential, the criterion was retained in the framework.

2.3. Methods to develop the need framework

Selected patient and societal needs criteria and corresponding indicators were categorised and reorganised into different dimensions and domains, to present them in the most meaningful way in relation to the research aim. This was a three iterations process involving experts of the research team and representatives of the main health system stakeholders in Belgium.

3. Results

3.1. Literature review on patient needs

From 478 screened papers, nine studies were selected (Figure A1, supplementary material). Notable tools include the Camberwell Assessment of Need Short Appraisal Version (CANSAS), which evaluates needs across multiple domains such as basic needs, health and social aspects. The Cancer survivors' Unmet needs (CasUn) tool and the 5-D Itch scale address specific needs of cancer survivors and those with pruritus [24,25]. Key domains identified in Maertens de Noordhout et al. (2022) [18] include physical, psychological, autonomy, social, healthcare accessibility, and spiritual needs. Additional relevant domains identified through this literature review included the professional awareness, patient acceptance in various life areas, and stigma [26].

3.2. Literature review and stakeholder consultations on societal needs

Of 2216 screened abstracts, 153 were reviewed in full, and 43 were included (Figure A2, supplementary material). Seven societal needs domains were identified: burden of disease, economic burden, equity, preventability, transmissibility, environmental impact, and future public health trends. Twenty-two criteria were retained from the long list of criteria extracted from the 43 included studies (Table A4, supplementary material).

Thirteen stakeholders provided written feedback on the relevance and essentiality of the proposed societal needs criteria. Twenty-two stakeholders attended the stakeholder meeting in April 2023 (Table A4). Through the combination of written and oral feedback, all relevant stakeholder groups in Belgium were represented. Among the 22 societal needs criteria retained, 10 did not reach the threshold of at least 60 % of participants reporting them as highly relevant, and >50 % reporting them as essential. These criteria were removed, except for the criterion "Greenhouse gas emissions related to standard of care" which was highlighted as important to keep by one stakeholder during the stakeholder meeting. As a result of this consultation, two new indicators were added to the "Environmental impact" dimension, i.e. "hazardous waste related to standard of care" and "use of natural resources and raw materials related to standard of care".

3.3. The need assessment framework

The NEED framework integrates three dimensions—patient, societal, and future needs—across three domains: health, healthcare, and social needs. Some of these needs are causally interconnected; for example, health needs can lead to healthcare needs. Nevertheless, the framework covers all three types of needs, as its aim is to highlight the principal existing areas of unmet needs. Each dimension and domain is evaluated using 23 criteria and 43 indicators to measure the extent of needs being met or unmet (Table 1, 2 and 3).

3.4. Dimensions

The framework focuses on health-related needs, and within this scope comprises three dimensions: patient, societal, and future needs. Patient needs result from the direct impact of a health condition on patients' lives and well-being. Societal needs encompass the broader impact of health conditions on society by considering the negative externalities of a disease. They take into account the interactions and interdependencies between individual needs and societal values, and include collective needs that extend beyond individual patient needs and affect the functioning and well-being of society as a whole [27]. The future needs dimension anticipates the long-term health and economic burdens that future generations might face, aiming to identify interventions to mitigate these impacts. The future needs dimension was included as an additional dimension on the framework after internal and

Table 1

Criteria, indicators and suggested data collection methods to identify and measure patient health-related needs.

Domain	Criterion	Indicator	Indicator description	Suggested data collection method(s)*
Health	Impact on general health-related quality of life	Years of life lived with disability (per case)	Number of years of life lost due to disability or ill-health caused by the health condition per patient.	Databases analysis
		EQ-5D-5 L average score	Score measuring health-related quality of life along five dimensions: mobility, self-care, usual activities, pain/discomfort, anxiety/depression.	Patient survey
	Impact on physical health	Associated comorbidities	Names of associated comorbidities and their frequencies.	Literature review
		Experienced burden of physical symptoms	Burdensome level of physical symptoms experienced by the patients due to the health condition.	Patient survey and/or interviews
		Pain/discomfort today vs before	Difference in the level of pain or discomfort today compared to before the onset of the health condition.	Patient survey and/or interviews
	Impact on psychological health	Experienced burden of psychological symptoms	Burdensome level of psychological symptoms experienced by the patients due to the health condition.	Patient survey and/or interviews
		Anxiety/depression today vs before	Difference in the level of anxiety or depression today compared to before the onset of the health condition.	Patient survey and/or interviews
	Impact on autonomy	Mobility today vs before onset of disease	Difference in the extent of mobility problems today compared to before the onset of the health condition.	Patient survey and/or interviews
		Self-care today vs before onset of disease	Difference in the extent of self-care problems today compared to before the onset of the health condition.	Patient survey and/or interviews
		Usual activities before vs today	Difference in the extent of problems with doing usual activities today compared to before the onset of the health condition.	Patient survey and/or interviews
Healthcare	Impact on life expectancy	Years of life lost (per death)	Number of years of life lost due to the health condition per death.	Databases analysis
	Treatment effectiveness	Effectiveness of current treatment	Overall effectiveness level of the current treatment or standard of care.	Expert opinion
	Burden of treatment	Experienced burden of treatment side-effects	Burdensome level of side-effects of the current treatment for the health condition.	Patient survey and/or interviews
		Other experienced burden of treatment	Burdensome level of the current treatment for the health condition (other than side-effects). (E.g. the administration route, the necessity to travel to hospital for treatment, the frequency of administration, or restrictions related to the storage of treatment).	Patient survey and/or interviews
	Quality of care	Experience with the organisation of care	Extent to which healthcare is organised in a way that works for patients.	Patient survey and/or interviews
		Quantity of information received	Amount of explanation patients received from their caregivers about their health condition and its treatment.	Patient survey and/or interviews
		Clarity of information received	Clarity of explanation patients received from their caregivers about their health condition and its treatment.	Patient survey and/or interviews
		Involvement in treatment choice	Extent to which patients would like or would have liked to be more involved in the choices about the treatment for their health condition.	Patient survey and/or interviews
		Diagnostic timing	Amount of time that has passed between the patient's first medical appointment and the diagnosis of their health condition.	Patient survey and/or interviews
	Accessibility of healthcare	Treatment availability	Extent of availability problems related to the current treatment, including problems related to both the logistic and financial accessibility of the treatment.	Expert opinion
		Forgone care	Not using health care despite perceiving a need for it, which can be due to several reasons, including costs, waiting lists or travel difficulties.	Patient survey and/or interviews
Social	Impact on social life	Social support needs	Extent to which patients needed social support because of their health condition without reaching it.	Patient survey and/or interviews
	Impact on education	Years of education lost	Amount of time lost in education due to the health condition.	Patient survey and/or interviews
	Impact on work	Changes in working conditions	Extent to which the health condition changed patients' ability to remain professionally active, considering factors such as working hours.	Patient survey and/or interviews
	Financial consequences	Average out-of-pocket healthcare costs per patient	Annual average amount of money a patient pays for healthcare costs because of the health condition that are not covered by health insurance.	Databases analysis and patient interviews
		Average out-of-pocket non-healthcare costs per patient	Annual average amount of money a patient pays for non-healthcare costs because of the health condition that are not covered by health insurance.	Patient interviews

* Literature review as a data collection method is applicable to all indicators. The recommended data collection methods for each health condition might vary based on the quality of data available from the different sources.

external discussions with stakeholders in order to highlight that a health issue might generate unmet health-related patient and societal needs in the future that differ from the present ones. Indeed, the impact that a health condition will have on patients and society in the future (due to demographic changes, climate change, emerging diseases, changing risk factors, antimicrobial resistance etc.) must also be taken into account to signal the urgency of tackling a given health condition.

Equity is also integrated into the framework as a transverse dimension, because it affects all dimensions. Equity implies the absence of unfair and avoidable or remediable differences in health, healthcare or social aspects among population groups defined socially, economically, demographically, geographically or by other factors. The health-related needs associated with a health condition can vary within a patient population, influenced by factors such as age, gender, or socioeconomic status. To identify potential inequities, unmet needs data should be analysed by relevant sub-group population.

3.5. Domains

Health conditions generate health, healthcare, and social needs. In the framework, these are referred to as domains. Health needs are needs for better health (e.g. in case of illness, disability, or injury) and highlight the burden of the disease, including how much pain, suffering, reduction in wellbeing and quality of life the health condition is causing to patients or society. Healthcare needs are needs for healthcare services in a broad sense, including medical treatments but also preventive measures, nursing services, etc. Medical needs, a term often used in pharmaceutical policy, refer to the need for a (better) treatment and we therefore consider this to be a specific example of a healthcare need [28]. Social needs pertain to aspects of the socio-economic environment, such as social support, work or the environmental footprint of healthcare.

3.6. Criteria and indicators

Across the three domains, thirteen patient needs criteria were identified. Patient *health* needs encompass five criteria. “Impact on general health-related quality of life (HRQoL)” assesses the need for improved overall HRQoL by means of indicators on the number of years lived with disability (YLD per case), the EQ-5D-5 L average utility score and the frequency of comorbidities associated with the health condition. The latter indicator may help to accurately interpret the difference in general HRQoL before the onset of disease and HRQoL today, and may point towards unmet needs linked to comorbidities which are not captured by disease-specific indicators. “Impact on physical health” captures the need for better physical health by measuring individuals’ experienced burden of physical symptoms given current standard of care, and the change in pain or discomfort before and after the onset of the health condition. “Impact on psychological health” captures the need for better psychological health by measuring individuals’ experienced burden of psychological symptoms given current standard of care, and change in anxiety or depression before and after the onset of the health condition. “Impact on autonomy” assesses the need for improved autonomy by means of changes in self-reported mobility, self-care, and usual activities before versus after the onset of the health condition. “Impact on life expectancy” identifies the need to maintain the average life expectancy and is measured using the number of years of life lost (YLL per death) of individuals living with a health condition.

Patient *healthcare* needs can be identified using four criteria. “Effectiveness of current treatment” identifies the need for an effective treatment for the patient population. The overall treatment for a health condition is considered to be very effective if it is curative or disease-modifying and patients can achieve a very good quality of life without complications. The criterion “Burden of current treatment” captures the need for a treatment that is manageable by measuring the experienced burden and side-effects of treatment. “Quality of care” detects the need

for high quality of care, which encompasses indicators related to the experience with the organisation of care, the quantity and clarity of information received from healthcare professionals, patients’ involvement in treatment choices and the timing of the diagnosis. The “Accessibility” criterion focuses on the need to have easy access to treatments and healthcare services, which are measured by indicators for the availability of treatment (i.e. logistic and financial accessibility of treatment) and foregone care. Together, indicators related to the availability, effectiveness, and burden of current treatment highlight potential unmet therapeutic needs.

Four criteria can be used to identify patient *social* needs. “Impact on social life” identifies the need for improved social life, including better or increased logistical, administrative, spiritual or religious support or social help than currently provided and more frequent or meaningful interactions with the community, family and/or friends. “Impact on education” detects educational needs, by measuring the amount of time lost in education. “Impact on work” highlights needs related to work and the working environment, by measuring changes in working capacity before and after the onset of the health condition. “Financial consequences” discerns patients’ financial needs resulting from the health condition, which can be assessed by the average out-of-pocket healthcare costs and non-healthcare costs per patient caused by the health condition.

Eight societal needs criteria were identified. Societal *health* needs can be identified through four criteria. “Frequency” highlights the need to decrease the prevalence or incidence of a health condition in the population. “Transmissibility” identifies the need to control or lower the transmission of a disease from one organism to another. Combined with the indicators of patient health needs, these two indicators reflect the societal disease burden of a health condition. The criterion “antimicrobial resistance (AMR)” specifies the need for society to tackle AMR related to the standard of care for a specific health condition. As AMR poses a significant threat to public health and has broad societal implications, it is important to consider and anticipate the disease-specific unmet needs related to AMR. The need to address AMR is assessed by the current presence of resistance to treatments used and, looking forward, by monitoring the volume of antibiotics delivered to treat a health condition. “Health burden on informal caregivers” identifies the need for improving informal caregivers HRQoL, including physical health, psychological health, and autonomy.

Societal *healthcare* needs can be identified through two criteria. “Value for money of standard of care” identifies inefficiencies in the healthcare system, which lead to opportunity costs for society and thus create unmet needs from a societal perspective. “Preventability” captures the societal need for preventing a specific health condition through primary prevention whenever possible. Primary prevention strategies should consistently be considered as potential solutions, together with information on their feasibility, costs, effectiveness, and other benefits. Secondary prevention measures (e.g. screening programmes) are included in our definition of treatment and therefore covered by the patient healthcare needs criteria (e.g., availability and effectiveness of treatment).

Societal *social* needs can be identified by two criteria. “Productivity losses” identifies the need to reduce the economic impact that a health condition has on society, measured by productivity losses due to absenteeism, presenteeism, and early labour force exits (e.g., retirement) of patients and their informal caregivers. “Environmental impact of standard of care” detects the need for a more environmentally sustainable standard of care or treatment, encompassing the amount of greenhouse gas emissions, hazardous waste and use of natural resources and raw materials related to the standard of care.

Two future needs criteria were identified, including future burden of disease and future economic burden. “Future burden of disease” identifies the need to reduce the potential health burden of a health condition on future patients and society, by estimating its future frequency, total years of life lost and total years lived with disability due to the

Table 2

Criteria, indicators and suggested data collection methods to identify and measure societal health-related needs.

Domain	Criterion	Indicator	Indicator description	Suggested data collection method (s)*
Health	Frequency	Prevalence or incidence	Prevalence: the total number of individuals in the population who have the health condition at a specific period of time. Incidence: the number of individuals who develop the health condition during a particular time period.	Database analysis
	Transmissibility	Contagiousness	Extent to which a health condition can be directly or indirectly transmitted from one organism to another.	Literature review
	Antimicrobial resistance	Existing antimicrobial resistance to treatments used	Extent to which antimicrobial resistance exists to the current treatment used for the health condition.	Expert Opinion
		Volume of antibiotics delivered to treat the disease	Amount of antibiotics prescribed to patients as part of the current treatment for the health condition.	Databases analysis
	Burden on informal caregivers	Loss in informal caregivers' quality of life	Extent to which the health condition negatively impacts the health-related quality of life of informal caregivers.	Literature review
Healthcare	Value for money of standard of care	Healthcare costs per unit of health gain	Disaggregated presentation of total healthcare expenditures and total impact of the current standard of care for the health condition on health outcomes.	Databases analysis
	Preventability	Implementation of effective primary prevention strategies	Extent to which effective primary prevention strategies are implemented for the given health condition.	Expert opinion
Social	Productivity losses	Costs related to work absenteeism, presenteeism, or/and early labor force exits of patients	Amount of costs due to the absence of patients from their workplace (i.e. time taken off work), lost productivity that occurs when patients are not fully functioning in the workplace or exit the labor force earlier because of their health condition.	Databases analysis
		Costs related to work absenteeism, presenteeism or/and early labor force exits of caregivers	Amount of costs due to the absence of informal caregivers from their workplace (i.e. time taken off work), lost productivity that occurs when informal caregivers are not fully functioning in the workplace or exit the labor force earlier because of their caregiving activities.	Literature review
	Environmental impact of standard of care	Greenhouse gas emissions related to the standard of care	Estimated amount of direct greenhouse gas emissions (scope 1: emitted by healthcare providers) and indirect greenhouse gas emission (scope 2: derived from primary resources and scope 3: derived from products used by healthcare providers) related to the standard of care for the health condition.	Expert opinion
		Hazardous waste related to standard of care	Estimated volume of hazardous (e.g. radioactive, infectious) waste generated as part of the standard of care for the health condition.	Expert opinion
		Use of natural resources and raw materials related to standard of care	Estimated volume of natural resources and raw materials utilised in the standard of care for the health condition.	Expert opinion

* Literature review as a data collection method is applicable to all indicators. The recommended data collection methods for each health condition might vary based on the quality of data available from the different sources.

health condition. "Future economic burden" discerns the need to reduce the potential economic burden of a health condition on future patients and society, by measuring future direct and indirect costs.

3.7. Data collection and measurement methods for populating the framework

To populate the NEED assessment framework with data on the health-related needs criteria and indicators, a mixed-method approach combining both primary and secondary data collection methods is recommended (Tables 1, 2 and 3). Primary data collection methods include a patient survey, individual patient interviews and expert opinions. Patient involvement throughout the primary data collection process is key and it is essential to consider at which stage and how patients can be involved. Secondary data collection methods include reviews of the scientific and grey literature, and the analysis of data from existing databases. For each health-related need indicator, more information on the suggested measurement methods (e.g. survey question) can be found in Tables A7 and A8 (supplementary material).

4. Discussion

This study introduces the NEED framework, a novel approach to assessing unmet health-related needs from patient and societal

perspectives. It evaluates needs across three dimensions: patient, societal, and future, and includes equity as a key aspect. Each dimension covers health, healthcare, and social needs, with 23 criteria and 43 indicators to measure them.

Criteria and indicators in the NEED framework could be used in the priority-setting processes of health(care) system decision makers. For each indicator, threshold values will have to be defined to determine the level of unmet needs. The relative importance of different needs criteria will have to be decided by users of the framework depending on their objective.

At the EU-level, the NEED framework can contribute to the discussions about the definition of the term "unmet medical needs" in the EU pharmaceutical legislation and the health technology assessment (HTA) regulation [5,29]. If in the revised EU pharmaceutical legislation incentives are foreseen for the development of medicinal products that address an unmet medical need or a high unmet medical need for rare diseases, it should be clear which criteria will be considered in the appraisal of unmet medical needs. The implications of the definition of "unmet medical needs" will likely go beyond R&D and affect the post-authorisation phase and HTA. Therefore, pricing and reimbursement (P&R) agencies, national and multi-national HTA agencies and medicines agencies, which are key potential users of this framework, could use the patient and societal need criteria to evaluate the level of unmet needs for decision-making. R&D funding organisations could also use criteria in the framework to evaluate health-related research

Table 3

Criteria, indicators and suggested data collection methods to identify and measure future health-related needs.

Domain	Criterion	Indicator	Indicator description	Suggested data collection method(s) [*]
Health	Future burden of disease	Future prevalence/incidence	Future prevalence: total number of individuals in the population who have the health condition at a specific period of time in the future. Future incidence: number of individuals who develop the health condition during a particular time period in the future.	Databases analysis
		Future years of life lost	Future number of years of life lost due to the health condition.	Databases analysis
		Future years lived with disability	Future number of healthy years of life lost due to disability or ill-health caused by the health condition.	Databases analysis
Healthcare and social	Future economic burden	Future direct costs	Amount of healthcare costs related to the management of the health condition in future years.	Databases analysis
		Future indirect costs	Amount of costs incurred from the cessation or reduction of work productivity as a result of the health condition in future years.	Databases analysis

^{*} Literature review as a data collection method is applicable to all indicators. The recommended data collection methods for each health condition might vary based on the quality of data available from the different sources.

proposals and allocate research funding. Furthermore, the NEED framework could contribute to the discussions on assessing the importance and feasibility of using unmet need for healthcare services as an additional indicator to monitor universal health coverage nationally and globally, as requested in one of the resolutions of the 76th World Health Assembly [30,31].

The NEED assessment framework is part of the larger NEED approach which aims to create an evidence database on patient and societal unmet needs [19]. Such a database could be used by P&R agencies, medicines agencies, HTA bodies, health and social care ministries, researchers, research funders, health product developers, patient organisations, and healthcare providers to prioritise areas for research, for the development of interventions, and to assess the extent to which proposed 'solutions' meet the most pressing unmet needs of patients or society.

This study offers a comprehensive definition of the concept of "health-related needs", which encompasses unmet needs from both the patient and the wider societal perspective. The concept of unmet patient needs defined in this study goes beyond unmet medical or therapeutic needs and also covers unmet healthcare and social needs. This expansion

is important to ensure patient-centred, value-based and high-quality patient care. At the societal level, this study is the first to present a set of criteria and indicators to identify and measure societal needs. By including criteria to identify the future needs generated by a health condition, the NEED framework adopts a long-term perspective, susceptible to recognising future public health threats. Overall, the synthesis of literature findings, stakeholder insights, and expert discussions culminated in a robust framework containing criteria and indicators for assessing the unmet needs of patients and society.

Nevertheless, the NEED framework has limitations. First, using a disease-based approach may lead to overlooking crucial health-related needs that are transverse to various diseases. While these unmet needs may contribute only little to each specific disease, taken as a whole, they are or should be considered public health priorities. For instance, environmental risk factors, and the social and commercial determinants of health are responsible for a high burden of disease through their contribution to numerous health conditions, warranting recognition as pressing societal unmet needs [32]. We should also acknowledge that needs are positively associated with the presence of multimorbidity and that this problem will only become more acute in the coming years with an ageing population. A disease-based approach cannot fully take this aspect into account. However, to consider this limitation as best we can, the NEED assessment framework includes an indicator on the frequency of co-morbidities associated with each health condition. This information may help to interpret other indicators, e.g. the difference in general HRQoL before the onset of disease and today, and may highlight the existence of additional (unmeasured) unmet needs linked to comorbidities that are not captured by disease-specific indicators. Yet, when assessing the contribution of the risk factor(s) to individual conditions in isolation, their significance may not be very high. Other transverse issues, such as unmet needs within the health system (e.g. insufficient or strained workforce) or research gaps not specific to a particular disease (e.g. the underrepresentation of pregnant women in clinical trials) might also be overlooked when using a disease-based approach for health (care) policy prioritisation. It is therefore crucial to emphasise that transverse health-related unmet needs must be separately considered and assessed. While the inclusion of a criterion for primary prevention in our framework might, at least partly, address these risk factors, it may not sufficiently emphasise the potential of mitigating upstream risk factors. A disease-based approach inherently hinges on an existing diagnosis. Consequently, medical conditions which have not yet been recognised under a specific name, cannot be assessed by this framework. Patients affected by such conditions likely experience significant unmet health-related needs, given the uncertainty, often coupled with an underappreciation of their suffering and a lack of tailored healthcare solutions. Second, the NEED framework has methodological limitations due to inherent issues within the indicators it utilises. For instance, the indicator "YLD per case" relies on disability weights, which may be influenced by the selection of the panel composition chosen for evaluating health states in pairwise comparisons [33]. However, this limitation is mitigated in the framework by the inclusion of multiple indicators, which provides a more comprehensive assessment of the burden of disease than relying on one or few measures. Third, while we have tried to minimize overlap between needs specific criteria and indicators, the criteria are not mutually exclusive and some overlaps remain (e.g. experienced burden of physical symptoms and pain/discomfort today vs before), which could be a potential for overestimating certain unmet needs. Fourth, given the evolving nature of the framework, some criteria or indicators may be missing, such as those to identify research needs, or incomplete, such as the criterion of "quality of care," and some may currently be difficult to assess due to a lack of data or measurement method. Finally, the applicability of the NEED framework might have to be adapted for specific health conditions (e.g. rare diseases, acute conditions), vulnerable population groups (e.g. children), or specific situations (e.g. public health crisis). The framework should also be validated at the European level and a reduced

version might increase its ease of use for policy-makers.

5. Conclusion

The NEED assessment framework provides a structured comprehensive approach to evaluate health-related needs. By considering patient, societal, and future perspectives, the framework offers valuable insights for informing healthcare policies, ultimately aiming to improve population health and well-being.

5.1. Policy recommendations

- Consider using the NEED assessment framework to inform health policy decisions and allocation of research funding.
- Continue to encourage and promote needs-based healthcare policies, based on the identification of the needs of patients and society using sound and transparent scientific approaches.

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CRediT authorship contribution statement

Charline Maertens de Noordhout: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Muriel Levy:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Rani Claerman:** Writing – review & editing, Methodology, Formal analysis. **Mats de Jaeger:** Writing – review & editing, Formal analysis. **Robby De Pauw:** Writing – review & editing, Validation, Formal analysis, Conceptualization. **Laurence Kohn:** Writing – review & editing, Validation, Methodology, Formal analysis, Conceptualization. **Claudia Schönborn:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Irina Cleemput:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declarations of competing interest

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Supplementary materials

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